



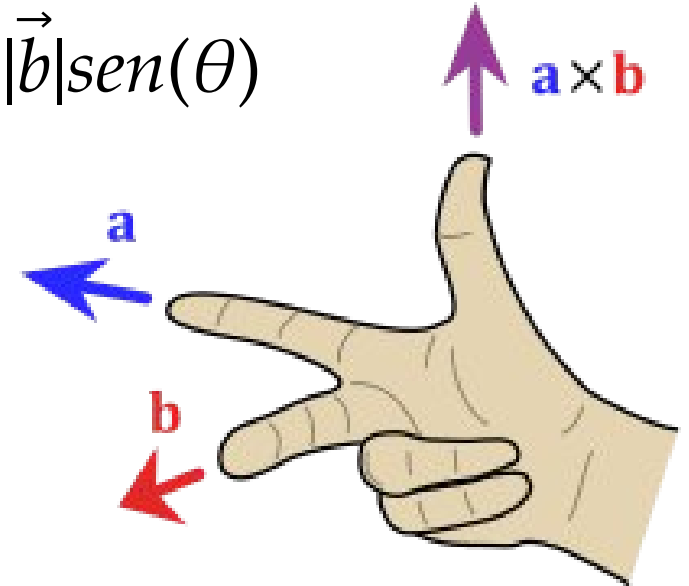
1 – Estática da Partícula / Corpo Rígido

Vetores (Produto vetorial)

$$\vec{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = (a_1, a_2, a_3) = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}$$

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin(\theta)$$

$$\vec{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = (b_1, b_2, b_3) = b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k}$$



$$\vec{a} \times \vec{b} = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ \mathbf{i} & \mathbf{j} & \mathbf{k} \end{pmatrix} = (a_2 b_3 - a_3 b_2) \mathbf{i} + (a_3 b_1 - a_1 b_3) \mathbf{j} + (a_1 b_2 - a_2 b_1) \mathbf{k}$$